

□ **SLOT GAME DEVELOPER MASTER GUIDE**

Complete Feature Guide with PixiJS Code

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1. Slot Game — Basic Concepts

1.1 Grid Structure

Slot game ek grid hoti hai — Reels (columns) aur Rows (lines) se bani. Standard grid 5x3 hoti hai — 5 reels, 3 rows.

Term	Meaning	Example
Reel	Vertical column — symbols spin karte hain	5 reels = 5 columns
Row	Horizontal line — symbols display hote hain	3 rows = 3 lines
Symbol	Game ka character — A, K, Q, Wild, Scatter	7, Bell, Cherry
Payline	Win hone ka pattern — symbols ka sequence	Left se right, zigzag
Bet	Player kitna lagata hai	₹1, ₹10, ₹100
Win	Payline match hone pe milne wala amount	Bet x Multiplier

1.2 Symbols Types

Symbol Type	Kya Karta Hai	Priority
Regular Symbol	Normal win deta hai — A, K, Q, J, 10	Normal
Wild Symbol	Kisi bhi symbol ki jagah le sakta hai	High
Scatter Symbol	Bonus/Free Spins trigger karta hai — anywhere land ho sakta hai	Highest
Bonus Symbol	Mini game ya special feature trigger karta hai	Special
Mystery Symbol	Land hone ke baad koi random symbol ban jaata hai	Special

1.3 Basic Flow

Ek spin ka basic flow:

- Player bet lagata hai

- Spin button dabata hai
- Reels spin karti hain (animation)
- Reels ruk jaati hain — result aata hai
- Paylines check hoti hain — win calculate hoti hai
- Win animation play hoti hai
- Balance update hota hai
- Next spin ke liye ready

2. Wild Features — Complete Guide

2.1 Standard Wild

Sabse basic wild — kisi bhi regular symbol ki jagah le sakta hai. Scatter ki jagah nahi le sakta.

Logic:

- Win check karte waqt — agar wild aaya toh usse kisi bhi symbol ki jagah consider karo
- Highest win wala combination choose karo

Code:

```
function isWild(symbol) {  
    return symbol === 'WILD';  
}  
  
function checkSymbolMatch(sym1, sym2) {  
    // Wild kisi se bhi match karta hai  
    if (isWild(sym1) || isWild(sym2)) return true;  
    return sym1 === sym2;  
}
```

2.2 Random Wild

Spin ke baad — kuch random positions pe wild drop hota hai. Pehle result aata hai, phir wild add hota hai, phir win calculate hota hai.

Logic:

- Spin complete hone ke baad
- Random positions select karo (1 to 5 positions)
- Un positions ke symbols Wild se replace karo
- Phir win calculate karo

Code:

```
function addRandomWilds(grid, count = 3) {  
    const positions = [];  
    const allPositions = [];  
  
    // Saari possible positions collect karo  
    for (let reel = 0; reel < grid.length; reel++) {
```

```

    for (let row = 0; row < grid[reel].length; row++) {
      allPositions.push({ reel, row });
    }
  }

  // Random positions choose karo
  for (let i = 0; i < count; i++) {
    const randomIndex = Math.floor(Math.random() * allPositions.length);
    positions.push(allPositions.splice(randomIndex, 1)[0]);
  }

  // Grid update karo
  positions.forEach(pos => {
    grid[pos.reel][pos.row] = 'WILD';
  });

  return positions; // Animation ke liye positions return karo
}

```

2.3 Expanding Wild

Wild land hota hai aur poori reel (column) wild ban jaati hai. Pehle original wild glow karta hai, phir baaki cells expand hoti hain.

Logic:

- Spin result check karo — kisi reel mein wild hai?
- Agar hai — us reel ki saari cells wild kar do
- Multiple wilds ek reel mein — koi farak nahi, poori reel wild hogi
- Animation: pehle original wild glow → phir expand
- Phir win calculate karo

Code:

```

function checkExpandingWilds(grid) {
  const expandedReels = [];

  grid.forEach((reel, reelIndex) => {
    // Sare wild positions dhundo
    const wildPositions = reel
      .map((sym, i) => sym === 'WILD' ? i : -1)
      .filter(i => i !== -1);

    if (wildPositions.length > 0) {
      // Poori reel wild karo
      for (let row = 0; row < reel.length; row++) {

```

```

        reel[row] = 'WILD';
    }
    expandedReels.push({ reelIndex, wildPositions });
}
});

return expandedReels;
}

```

□ *Expanding wild ke baad hi win calculate karo — pehle nahi!*

2.4 Sticky Wild

Wild ek jagah fix rehta hai multiple spins tak. Free spins mein use hota hai. Wild land hua → us position pe locked → baaki reels spin karti hain.

Logic:

- Wild land hua → position track karo
- Next spin pe — us position ko spin mat karo
- Wild wahan hi rehta hai
- Counter track karo — kitne spins baad hatega

Code:

```

this.stickyWilds = []; // { reel, row, spinsLeft }

function addStickyWild(reel, row, spins = 3) {
    this.stickyWilds.push({ reel, row, spinsLeft: spins });
}

function applyStickyWilds(grid) {
    this.stickyWilds.forEach(wild => {
        grid[wild.reel][wild.row] = 'WILD';
    });
}

function updateStickyWilds() {
    // Har spin ke baad countdown karo
    this.stickyWilds = this.stickyWilds
        .map(w => ({ ...w, spinsLeft: w.spinsLeft - 1 }))
        .filter(w => w.spinsLeft > 0); // Expire hue wild hata do
}

```

2.5 Walking Wild

Wild land hota hai aur har spin pe ek position move karta hai — right se left — jab tak screen se bahar na chala jaye.

Code:

```
this.walkingWilds = []; // { reel, row }

function moveWalkingWilds() {
  this.walkingWilds = this.walkingWilds
    .map(w => ({ ...w, reel: w.reel - 1 })) // Left move
    .filter(w => w.reel >= 0); // Screen se bahar gaye to remove
}
```

2.6 Multiplier Wild

Wild ke saath multiplier attached hota hai — 2x, 3x, 5x. Win mein wo multiplier apply hota hai.

Code:

```
const wildMultipliers = { 'WILD_2X': 2, 'WILD_3X': 3, 'WILD_5X': 5 };

function getWildMultiplier(symbol) {
  return wildMultipliers[symbol] || 1;
}

function calculateWinWithMultiplier(baseWin, wilds) {
  let multiplier = 1;
  wilds.forEach(sym => multiplier *= getWildMultiplier(sym));
  return baseWin * multiplier;
}
```

3. Cascading / Tumble Reels

3.1 Concept

Winning symbols hat jaate hain → upar se naye girte hain → phir win check hota hai → chain chalti rehti hai jab tak koi win na aaye.
Isko Chain Reaction ya Avalanche bhi bolte hain. Popular games: Gonzo's Quest, Sweet Bonanza.

3.2 Visual Example

Step	Grid State	Action
1. Spin	A B W A C / B A A D B / W A C B A	Normal spin result
2. Win Check	Row 2: A A A = WIN!	3 A symbols match
3. Remove	A B W A C / B _ _ D B / W A C B A	Winning A symbols hat gaye
4. Drop	Naye symbols upar se aate hain	Gravity effect
5. Check Again	Koi win hai?	Agar haan → repeat, nahi → stop

3.3 Complete Code

```
async function startCascade() {
  let cascadeCount = 0;
  this.totalWin = 0;

  while (true) {
    // Step 1: Win check karo
    const wins = checkWins(this.grid);

    if (wins.length === 0) break; // Koi win nahi - stop

    cascadeCount++;
    this.totalWin += calculateWinAmount(wins);

    // Step 2: Winning symbols mark karo
    const toRemove = getWinningPositions(wins);

    // Step 3: Animation - symbols glow aur hat jaayein
    await playRemoveAnimation(toRemove);
  }
}
```



```

// Step 4: Grid update karo
removeSymbols(this.grid, toRemove);
fillEmptySpaces(this.grid);

// Step 5: Drop animation
await playDropAnimation();

// Safety limit
if (cascadeCount >= 20) break;
}

// Sab khatam - win dikhao
showTotalWin(this.totalWin);
enableSpinButton();
}

```

3.4 Remove & Fill Logic

```

function removeSymbols(grid, toRemove) {
  toRemove.forEach(pos => {
    grid[pos.reel][pos.row] = null; // null = khaali
  });
}

function fillEmptySpaces(grid) {
  grid.forEach(reel => {
    // Nulls ko upar bhejo (gravity simulate karo)
    // [A, null, B] → [null, A, B]
    for (let row = reel.length - 1; row >= 0; row--) {
      if (reel[row] === null) {
        for (let above = row - 1; above >= 0; above--) {
          if (reel[above] !== null) {
            reel[row] = reel[above];
            reel[above] = null;
            break;
          }
        }
      }
    }
    // Baaki null positions = naye random symbols
    for (let row = 0; row < reel.length; row++) {
      if (reel[row] === null) reel[row] = getRandomSymbol();
    }
  });
}

```

□ *Cascade ke saath Progressive Multiplier bhi lagao — 1x, 2x, 3x badhta jaye!*

4. Hold & Spin Feature

4.1 Concept

Special symbols (coins/orbs) collect karo → 3 chances milte hain (respins) → har naya symbol aane pe counter reset → jab tak 3 chances khatam na ho → total win collect karo.

4.2 Visual Flow

Step	Action
Trigger	3+ special symbols land karo normal spin mein
Lock	Special symbols lock ho jaate hain — reels freeze
Respin	3 chances milte hain — baaki reels spin karti hain
New Symbol	Naya special symbol aaya → locked → counter reset to 3
Empty Spin	Koi naya symbol nahi → counter -1
End	Counter 0 → sab symbols ka total collect karo

4.3 Code

```
this.holdAndSpinState = {
  active: false,
  lockedSymbols: [], // { reel, row, value }
  spinsLeft: 3,
};

function triggerHoldAndSpin(grid) {
  const specialSymbols = findSpecialSymbols(grid);

  if (specialSymbols.length >= 3) {
    this.holdAndSpinState.active = true;
    this.holdAndSpinState.lockedSymbols = specialSymbols;
    this.holdAndSpinState.spinsLeft = 3;
    return true;
  }
  return false;
}

async function doHoldAndSpin() {
  while (this.holdAndSpinState.spinsLeft > 0) {
```

```
// Sirf unlocked reels spin karo
const newResult = spinUnlockedReels();

// Naye special symbols dhundo
const newSpecial = findNewSpecialSymbols(newResult);

if (newSpecial.length > 0) {
  // Naye symbols lock karo
  newSpecial.forEach(s => this.holdAndSpinState.lockedSymbols.push(s));
  this.holdAndSpinState.spinsLeft = 3; // Reset!
} else {
  this.holdAndSpinState.spinsLeft--;
}

await playRespin Animation();
}

// Total collect karo
const total = this.holdAndSpinState.lockedSymbols
  .reduce((sum, s) => sum + s.value, 0);

showWin(total);
this.holdAndSpinState.active = false;
}
```

5. Free Spins Feature

5.1 Concept

3+ Scatter symbols aane pe Free Spins trigger hota hai. Player ko fixed number of free spins milte hain — bet nahi lagta. Usually special features bhi active hoti hain free spins mein.

5.2 Types

Type	Description
Basic Free Spins	Fixed spins — 10, 15, 20
Retriggerable	Free spins mein phir scatter aaye to aur spins milte hain
Multiplier Free Spins	Har spin pe multiplier badhta hai 1x → 2x → 3x
Sticky Wild Free Spins	Wilds sticky ho jaate hain free spins mein
Expanding Wild Free Spins	Wilds expand karte hain free spins mein

5.3 Code

```
this.freeSpinsState = {
  active: false,
  spinsLeft: 0,
  totalSpins: 0,
  multiplier: 1,
};

function checkScatterTrigger(grid) {
  let scatterCount = 0;

  grid.forEach(reel => {
    reel.forEach(symbol => {
      if (symbol === 'SCATTER') scatterCount++;
    });
  });

  // 3 scatter = 10 spins, 4 = 15, 5 = 20
  const spinsMap = { 3: 10, 4: 15, 5: 20 };
  return spinsMap[scatterCount] || 0;
}

async function startFreeSpins(spinsCount) {
```

```
this.freeSpinsState = {
  active: true,
  spinsLeft: spinsCount,
  totalSpins: spinsCount,
  multiplier: 1,
};

while (this.freeSpinsState.spinsLeft > 0) {
  this.freeSpinsState.spinsLeft--;

  // Normal spin logic - but free hai
  await doSpin();

  // Retrigger check
  const retrigger = checkScatterTrigger(this.grid);
  if (retrigger > 0) {
    this.freeSpinsState.spinsLeft += retrigger;
  }
}

this.freeSpinsState.active = false;
showFreeSpinsComplete();
}
```

6. Multiplier Features

6.1 Types of Multipliers

Type	Kaise Kaam Karta Hai	Where Used
Fixed Multiplier	Har win pe same — 2x ya 3x	Base game
Progressive Multiplier	Har cascade/tumble pe badhta hai	Cascade games
Scatter Multiplier	Jitne scatter utna multiplier	Scatter wins
Wild Multiplier	Wild pe multiplier attached	Special wilds
Free Spin Multiplier	Free spins mein badhta rehta hai	Free spins

6.2 Progressive Multiplier Code

```
this.cascadeMultiplier = 1;

async function startCascadeWithMultiplier() {
  this.cascadeMultiplier = 1;

  while (true) {
    const wins = checkWins(this.grid);
    if (wins.length === 0) break;

    // Multiplier apply karo
    const winAmount = calculateWinAmount(wins) * this.cascadeMultiplier;
    this.totalWin += winAmount;

    // Multiplier badhao
    this.cascadeMultiplier++;

    await removeAndDrop();
  }
}
```

7. Special Features

7.1 Mystery Symbol

Spin ke baad mystery symbols reveal hote hain — sab ek hi random symbol ban jaate hain. Win chances badhte hain.

Code:

```
function revealMysterySymbols(grid) {  
  // Random symbol choose karo (Wild aur Scatter nahi)  
  const symbols = ['A', 'K', 'Q', 'J', '10'];  
  const chosen = symbols[Math.floor(Math.random() * symbols.length)];  
  
  grid.forEach(reel => {  
    reel.forEach((sym, i) => {  
      if (sym === 'MYSTERY') reel[i] = chosen;  
    });  
  });  
}
```

7.2 Colossal Symbols (Giant Symbols)

2x2, 3x3 ya bade symbols ek saath land karte hain. Poora block same symbol hota hai — win chances bahut zyada hote hain.

Code:

```
function applyColossalSymbol(grid, startReel, startRow, size, symbol) {  
  for (let r = startReel; r < startReel + size; r++) {  
    for (let row = startRow; row < startRow + size; row++) {  
      if (grid[r] && grid[r][row] !== undefined) {  
        grid[r][row] = symbol;  
      }  
    }  
  }  
}
```

□ *Colossal symbol position check karo — boundary se bahar na jaye!*

7.3 Symbol Transformation (Butterfly Feature)

Ek ya zyada symbols transform ho jaate hain doosre mein. Usually low value symbols high value mein badte hain.

Code:

```
function transformSymbols(grid, fromSymbol, toSymbol, positions) {
  positions.forEach(pos => {
    if (grid[pos.reel][pos.row] === fromSymbol) {
      grid[pos.reel][pos.row] = toSymbol;
    }
  });
}
```

7.4 Symbol Collection

Har spin pe kuch symbols collect hote hain — meter bharne pe bonus trigger hota hai.

Code:

```
this.collectionMeter = 0;
this.collectionTarget = 20;

function collectSymbols(grid, targetSymbol) {
  grid.forEach(reel => {
    reel.forEach(sym => {
      if (sym === targetSymbol) {
        this.collectionMeter++;
      }
    });
  });

  if (this.collectionMeter >= this.collectionTarget) {
    triggerBonus();
    this.collectionMeter = 0; // Reset
  }
}
```

8. Jackpot System

8.1 Types

Jackpot Type	Description	Amount
Mini	Sabse chhota — frequently milta hai	Fixed small amount
Minor	Medium jackpot	Fixed medium amount
Major	Bada jackpot — kam milta hai	Fixed large amount
Grand/Mega	Sabse bada — bahut kam milta hai	Progressive / Fixed huge

8.2 Fixed Jackpot Code

```
const JACKPOTS = {
  MINI: 100,
  MINOR: 500,
  MAJOR: 5000,
  GRAND: 50000,
};

function checkJackpot(grid) {
  // Example: 5 jackpot symbols = grand jackpot
  const jackpotSymbols = grid.flat().filter(s => s === 'JACKPOT');

  if (jackpotSymbols.length >= 5) return 'GRAND';
  if (jackpotSymbols.length === 4) return 'MAJOR';
  if (jackpotSymbols.length === 3) return 'MINOR';
  if (jackpotSymbols.length === 2) return 'MINI';
  return null;
}
```

8.3 Progressive Jackpot

```
this.progressiveJackpot = 10000; // Starting amount

function addToJackpot(betAmount) {
  // Har spin ka 1% jackpot mein jaata hai
  this.progressiveJackpot += betAmount * 0.01;
}
```

9. Win Calculation

9.1 Payline Check

```
const PAYLINES = [
  [0, 0, 0, 0, 0], // Top row
  [1, 1, 1, 1, 1], // Middle row
  [2, 2, 2, 2, 2], // Bottom row
  [0, 1, 2, 1, 0], // V shape
  [2, 1, 0, 1, 2], // Inverted V
];

function checkWins(grid) {
  const wins = [];

  PAYLINES.forEach((payline, lineIndex) => {
    const symbols = payline.map((row, reel) => grid[reel][row]);
    const win = checkLineWin(symbols, lineIndex, payline);
    if (win) wins.push(win);
  });

  return wins;
}

function checkLineWin(symbols, lineIndex, payline) {
  const firstSymbol = symbols[0] === 'WILD' ? symbols[1] : symbols[0];
  let matchCount = 0;

  for (let i = 0; i < symbols.length; i++) {
    if (symbols[i] === firstSymbol || symbols[i] === 'WILD') {
      matchCount++;
    } else break;
  }

  if (matchCount >= 3) {
    const payout = PAYTABLE[firstSymbol][matchCount];
    return { lineIndex, symbol: firstSymbol, count: matchCount, payout,
      positions: payline.slice(0, matchCount).map((row, reel) => ({ reel, row })) };
  }
  return null;
}
```

10. PixiJS Animations — Common Patterns

10.1 Win Animation

```
async function playWinAnimation(positions) {
  const promises = positions.map(pos => {
    const sprite = this.symbolSprites[pos.reel][pos.row];

    return gsap.to(sprite.scale, {
      x: 1.2, y: 1.2,
      duration: 0.3,
      yoyo: true,
      repeat: 3,
      ease: 'power2.inOut'
    });
  });

  await Promise.all(promises);
}
```

10.2 Glow Effect

```
// @pixi/filter-glow install karo
import { GlowFilter } from '@pixi/filter-glow';

function addGlow(sprite, color = 0xFFD700) {
  sprite.filters = [new GlowFilter({
    distance: 15,
    outerStrength: 2,
    innerStrength: 1,
    color: color,
  })];
}

function removeGlow(sprite) {
  sprite.filters = [];
}
```

10.3 Counter Animation (Win Amount)

```
function animateCounter(from, to, duration = 1000) {
  const startTime = Date.now();

  return new Promise(resolve => {
```

```
function update() {
  const elapsed = Date.now() - startTime;
  const progress = Math.min(elapsed / duration, 1);

  // Easing
  const eased = 1 - Math.pow(1 - progress, 3);
  const current = Math.floor(from + (to - from) * eased);

  updateWinDisplay(current);

  if (progress < 1) requestAnimationFrame(update);
  else resolve();
}
update();
});
}
```

11. Feature Quick Reference

Feature	Trigger	Core Logic	Difficulty
Standard Wild	Automatic	Symbol replacement	Easy
Random Wild	Post spin	Random position selection	Easy
Expanding Wild	Wild lands	Full reel = Wild	Easy
Sticky Wild	Wild lands in free spins	Position locked for N spins	Medium
Walking Wild	Wild lands	Move 1 position per spin	Medium
Multiplier Wild	Wild lands	Win x multiplier	Easy
Cascading Reels	Any win	Remove → Drop → Check loop	Medium
Progressive Multiplier	Each cascade	Multiplier++ per cascade	Easy
Hold & Spin	3+ special symbols	Lock → Respin → Collect	Hard
Free Spins	3+ scatter	N free spins with special rules	Medium
Mystery Symbol	Random	All mystery = same symbol	Easy
Colossal Symbol	Random	NxN block = same symbol	Medium
Symbol Collection	Every spin	Meter fill → bonus trigger	Medium
Jackpot	Special symbols	Fixed/Progressive payout	Medium